

Effects of high-velocity resistance training on muscle function, muscle properties, and physical performance in individuals with hip osteoarthritis: a randomized controlled trial

Clinical Rehabilitation
2014, Vol 28(1) 48–58
© The Author(s) 2013
Reprints and permissions:
sagepub.co.uk/journalsPermissions.nav
DOI: 10.1177/0269215513492161
cre.sagepub.com



Yoshihiro Fukumoto^{1,2}, Hiroshige Tateuchi², Tome Ikezoe², Rui Tsukagoshi², Haruhiko Akiyama³, Kazutaka So³, Yutaka Kuroda³ and Noriaki Ichihashi²

Abstract

Objective: To investigate the effects of high-velocity resistance training on muscle function, muscle properties, and physical performance in patients with hip osteoarthritis by comparison with those of low-velocity resistance training.

Design: Single-blind randomized controlled trial.

Setting: Home-based exercise programmes.

Subjects: A total of 46 women with hip osteoarthritis were randomly assigned to the high-velocity ($n = 23$) or low-velocity ($n = 23$) training group.

Interventions: Both groups underwent an eight-week daily home-based resistance training programme using an elastic band. Exercises involved hip abduction, extension, and flexion and knee extension. Participants in the high-velocity group performed the concentric phase of each repetition as rapidly as possible and returned to the initial position eccentrically in 3 s. Participants in the low-velocity group performed both the concentric and eccentric phases in 3 s.

Main measures: The following outcome measures were evaluated: isometric muscle strength, muscle power, muscle thickness, muscle echo intensity, maximum walking speed, Timed Up and Go test, 3-minute walking test, Harris Hip Score, and hip pain.

Results: Decreases in the time for performing the Timed Up and Go test (mean changes: high-velocity group -0.46 s, low-velocity group -0.23 s) and echo intensity of the gluteus maximus (mean changes: high-velocity group -6.8 , low-velocity group -1.0) were significantly greater in the high-velocity group than

¹Faculty of Rehabilitation, Kobe Gakuin University, Kobe, Japan

²Human Health Sciences, Graduate School of Medicine, Kyoto University, Kyoto, Japan

³Department of Orthopaedic Surgery, Graduate School of Medicine, Kyoto University, Kyoto, Japan

Corresponding author:

Yoshihiro Fukumoto, Faculty of Rehabilitation, Kobe Gakuin University, 518 Arise, Ikawadani-cho, Nishi-ku, Kobe 651-2180, Japan.

Email: fukumoto@reha.kobegakuin.ac.jp

in the low-velocity group. No significant difference was observed in changes of other outcome measures between the groups.

Conclusion: This study revealed that high-velocity training for patients with hip osteoarthritis has partially a greater effect on muscle properties and physical performance than low-velocity training.

Keywords

Hip osteoarthritis, muscle power, physical performance, treatment velocity, high-velocity resistance training

Received: 12 October 2012; accepted: 6 May 2013

Introduction

Osteoarthritis is a common musculoskeletal disorder. Lower limb osteoarthritis leads to pain, loss of mobility and muscle function, restricted activities of daily living, and decreased quality of life. In the clinical guidelines for patients with osteoarthritis of the hip or knee, therapeutic exercise is recommended to improve overall functioning.^{1,2} In contrast to knee osteoarthritis, however, far less research has been conducted into the role of exercise in hip osteoarthritis. Therefore, the most effective exercise for patients with hip osteoarthritis remains unclear.

In recent years, muscle power has been found to contribute more to improvement in physical function than muscle strength.^{3,4} Therefore, muscle power is becoming the focus of many resistance training studies in older adults. Previous studies in older adults reported that high-velocity resistance training (power training) resulted in significant increases in muscle strength, muscle power, and whole muscle cross-sectional area.^{5,6} Some studies^{7,8} compared high-velocity resistance training and conventional low-velocity resistance training in older adults and found that high-velocity training resulted in significantly greater improvement in leg press power and physical function than low-velocity training. Even at low intensity levels, high-velocity training improves muscle function and physical function.^{8–10} Orr et al.¹⁰ showed that high-velocity power training at low resistance produced the greater improvement in balance performance compared with that at high resistance, suggesting that training velocity, but not intensity of resistance, may be a critical variable during high-velocity training.

High external resistance has been recommended in resistance training for older adults. Even for patients with osteoarthritis, high resistance was often used in resistance training. However, in patients with osteoarthritis, high external resistance could cause overload of the joint and aggravate symptoms such as pain, swelling, and inflammation. In fact, some authors have suggested that exercise for patients with osteoarthritis should not include high joint load.^{11,12} Therefore, low-resistance training may be preferable and safer for patients with osteoarthritis. High-velocity training for patients with osteoarthritis may result in greater improvements in the performance of functional tasks, even with low resistance. To our knowledge, however, no study has investigated the effect of high-velocity training at low resistance in patients with hip osteoarthritis.

In this study, low-intensity home-based exercise at high-velocity or low-velocity using elastic bands was evaluated in patients with hip osteoarthritis. The objective of this study was to investigate the effect of high-velocity training on muscle function, muscle properties, and physical performance in patients with hip osteoarthritis by comparison with those of low-velocity training. We hypothesized that high-velocity training would have a greater impact on muscle power and physical performance than low-velocity training.

Methods

This randomized control trial was conducted at the Graduate School of Medicine, Kyoto University in compliance with the Declaration of Helsinki. It was

approved by the ethics committee of Kyoto University Graduate School and the Faculty of Medicine. Written informed consent was obtained from all participants prior to randomization.

The study recruited women diagnosed with bilateral or unilateral hip osteoarthritis who were managed as outpatients at the Department of Orthopaedic Surgery, Kyoto University Hospital, Kyoto, Japan and who had received no prior physical therapy. In this study, only women were recruited because muscle and physical function in women is lower than that in men, and gender difference is a major confounding factor that must be controlled. Inclusion criteria were as follows: ability to walk with or without assistive devices; absence of dementia; ability to understand the informed consent procedure; and no history of total hip arthroplasty, stroke, Parkinson's disease, neurological gait disorders, neuromuscular disease, vestibular problems, or symptoms affecting the knees, ankles, or back. Muscle strength deficit was not an inclusion criterion because it could not be clearly defined as no optimal cut-off level for identifying the strength deficit has been established even in healthy subjects. Muscle strength in patients with hip osteoarthritis is lower than that in healthy individuals, even for those with mild osteoarthritis.¹³ Candidate study participants were screened by four orthopaedic surgeons. Two study examiners conducted the study enrolment procedure.

Disease severity was assessed using a system based on the classification of the Japanese Orthopaedic Association.¹⁴ This system defined four stages of hip osteoarthritis: the pre-osteoarthritis stage (morphological changes of the acetabulum and/or proximal femur related to osteoarthritis), initial stage (one or more osteoarthritis changes and possible narrowing of the joint space), advanced stage (partial loss of the joint space), and terminal stage (gross loss of the joint space). Either the more severe side, or the more painful side if both hips were equally severe, was selected for analyses. After recruitment, participants were randomly assigned to the high-velocity or low-velocity resistance training group using a stratified block randomization technique (four participants per block). The participants were stratified by age (<50 years

and ≥ 50 years) and disease severity (pre-osteoarthritis or initial stage and advanced or terminal stage). In each block, permuted block randomization was then achieved using random numbers generated by Excel (Microsoft, Redmond, WA, USA). This procedure was conducted by a study examiner. Participants, but not study examiners (i.e. outcome assessors and a physical therapist who prescribed the intervention programme), were blinded to group allocation.

For both groups, a physical therapist prescribed an eight-week daily home-based resistance training programme. Four exercises were performed on both lower limbs: hip abduction in supine position, hip extension in prone position, hip flexion in sitting position, and knee extension in sitting position. Each exercise was performed using Theraband (Hygenic Co., Akron, OH, USA), and its colour indicated its resistance level (yellow, red, green, or blue). Exercise loads (as indicated by band colour) in resistance training were set at a level that was perceived as 'somewhat hard' by the participants. Two sets of 10 repetitions of concentric and eccentric contractions through the full range of motion were performed for the first two weeks, and three sets of 10 repetitions were performed thereafter. In addition, participants in the high-velocity group were instructed to perform the concentric phase of each repetition as rapidly as possible and then return through the eccentric phase in 3 s. Participants in the low-velocity group performed both the concentric and eccentric phases in 3 s.

The physical therapist contacted the participants every two weeks by telephone, and exercise loads were increased when participants could perform the exercises comfortably. To assist in monitoring compliance, participants maintained training diaries. The compliance rate was deemed as 100% if the participant had completed a total of 56 days of exercise over eight weeks.

Various parameters were evaluated at baseline and eight weeks after the intervention. This included muscle function (isometric strength and muscle power), physical performance, clinical assessment, and muscle properties (muscle mass and composition). The primary outcome was physical performance, as this study investigated whether

high-velocity training is more effective than low-velocity training in improving physical performance in patients with hip osteoarthritis.

Maximal isometric strength of the hip abductors, extensors, flexors, and knee extensors was measured as described previously.^{15,16} A hand-held dynamometer (μ Tas F-1; Anima Co., Tokyo, Japan) was used for the measurement of hip muscles, and an isokinetic dynamometer (Myoret RZ-450; Kawasaki Heavy Industries Ltd, Tokyo, Japan) was used for the measurement of knee extensors. Isometric strength was measured twice with a 30-s rest interval, and the resulting maximal value (Newton, N), estimated lever arm length (m), and body weight (kg) were combined to determine the torque to body weight ratio (Nm/kg).

As a measure of muscle power, the stairs ascending test was performed. The participants were asked to walk up 10 steps, each with a height of 16.5 cm, as fast as they could safely manage without the use of handrails or aids of any form. Power (W/kg) was calculated according to the method of Suetta et al.¹⁷

Maximum walking speed¹⁸ was measured over a 10-m floor length. Participants were provided with 2 m to accelerate and decelerate before and after the test distance and were asked to walk as fast as they could. Two trials were performed, and the faster speed (m/s) was used for analyses. The Timed Up and Go test¹⁹ was performed using a chair that was 43 cm in height. The participants were asked to stand up from a sitting position, walk 3 m, turn, walk back, and sit down quickly and safely. The faster time (s) from two performances of the Timed Up and Go test was obtained and used for analyses. The 3-min walking²⁰ was performed using a 25-m course, with cones as turning points. The participants were asked to walk as fast and safely as possible between the cones for 3 min. Distance travelled (m) in 3 min was recorded.

Clinical assessment was performed using the Harris Hip Score²¹ from 0–100, with 100 indicating the highest level of function. The Harris Hip Score is a disease-specific index containing eight items representing pain, walking function, activities of daily living, and range of motion of the hip joint. In addition, the impact of hip pain on daily activity

during the previous week was evaluated on a 100-mm visual analogue scale.

Ultrasound images were obtained using a B-mode ultrasound imaging device (LOGIQ e; GE Healthcare UK Ltd, Chalfont, UK) with an 8-MHz linear array probe. Three muscles were examined after the practice in a previous study:²² the gluteus maximus, gluteus medius, and quadriceps femoris. All measurements were performed by the same investigator once for each muscle. Muscle thickness of each muscle was measured as an index of muscle mass using an electric caliper on frozen images, as described previously.^{16,22} In addition, echo intensity was assessed as a muscle composition index and was determined by computer-assisted 8-bit greyscale analysis using the standard histogram function in Adobe Photoshop Elements (Adobe Systems Inc., San Jose, CA, USA), as described previously.^{16,23} The mean echo intensity of the regions was expressed as a value between 0 (black) and 255 (white). Enhanced echo intensity has been shown to indicate changes in muscle composition owing to increased intramuscular fibrous and adipose tissues.²⁴ Ultrasound measurement of these muscles demonstrated high reliability for muscle thickness (intra-class correlation coefficients: 0.90–0.99) and echo intensity (0.87–0.91).^{16,22,23} In addition, ultrasound measurement was shown to correlate with magnetic resonance imaging and computed tomography in terms of muscle cross-sectional area and muscle composition.^{25–27}

Mann–Whitney *U*-tests were used to analyse differences between the groups at baseline. To examine the differences in the effects of the intervention between the two groups, changes from baseline to postintervention were compared between the groups using analysis of covariance. Age, height, weight, and outcome measures at baseline were used as covariates. Mean differences and 95% confidence intervals were reported. On-treatment analysis, but not intention-to-treat analysis, was conducted. Missing values for participants who completed the study were not compensated. Statistical significance was set at $p < 0.05$. Statistical analyses were performed using SPSS version 17.0 (SPSS Japan Inc., Tokyo, Japan).

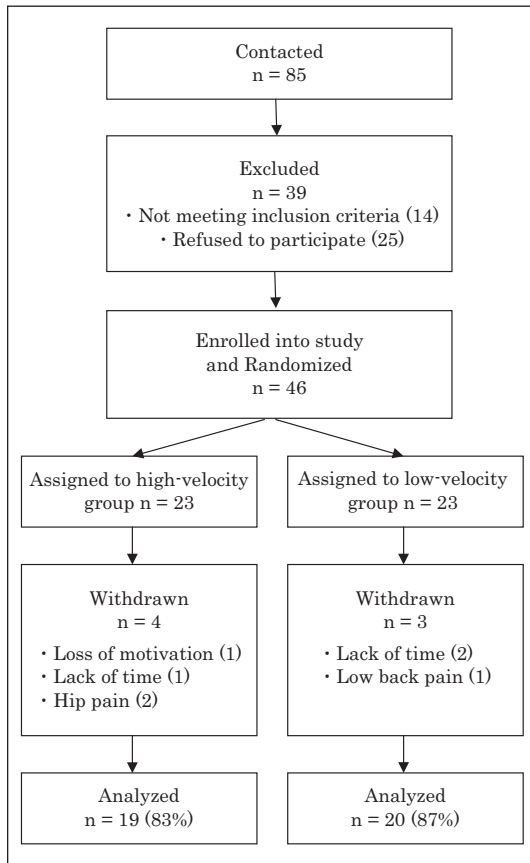


Figure 1. Flowchart of study participation.

Results

Between June 2010 and September 2011, 46 female subjects with a mean (SD) age of 53.4 (9.8) years were enrolled in this study. Figure 1 is a flowchart of study participation. Participants were randomly assigned to the high-velocity ($n = 23$) or low-velocity ($n = 23$) training group. Over the eight-week period of the study, seven participants (four high-velocity, three low-velocity) withdrew from the study. Of the four participants in the high-velocity group who withdrew from the study, one did so because of loss of motivation, one because of lack of time, and two because of hip pain during the exercise. In the low-velocity

group, two withdrew because of lack of time and one because of low back pain that was unrelated to the exercises. All other participants (high-velocity group, 19 participants and low-velocity group, 20 participants) completed the study. The mean (SD) compliance rate was 81% (23%) for the high-velocity group and 86% (13%) for the low-velocity group. No significant difference was observed between the groups.

Participant characteristics at baseline are shown in Table 1. For three participants (high-velocity group, two; low-velocity group, one), muscle power was not measured before or after the intervention because of their inability to ascend stairs without the use of handrails or aids. In addition, for one participant in the high-velocity group, muscle properties of the gluteus maximus and gluteus medius were not measured before or after the intervention because of refusal to undergo ultrasound. No significant differences were noted between the groups in physical characteristics, muscle function, muscle properties, physical performance, or clinical assessment.

Table 2 provides data on muscle function. No significant differences were noted between the groups in increases of all measurements of muscle strength or muscle power.

Table 3 shows the data on physical performance and clinical assessment. A significantly greater decrease in the time for performing the Timed Up and Go test was observed in the high-velocity group than in the low-velocity group ($p < 0.05$). No significant differences were observed between the groups in changes of maximum walking speed, 3-min walking test, Harris Hip Score, and pain measured on the visual analogue scale.

Table 4 shows the data on muscle properties. No significant difference was observed between the two groups in increase in muscle thickness for all muscles measured. With regard to echo intensity, a significant difference of decrease was observed for the gluteus maximus; this decrease was greater in the high-velocity group ($p < 0.05$). No significant differences were observed in decreases in the echo intensity of the gluteus medius and quadriceps femoris muscles.

Table 1. Characteristics of participants in the high-velocity ($n = 19$) and low-velocity groups ($n = 20$) at baseline.

Characteristic	High-velocity group ($n = 19$)	Low-velocity group ($n = 20$)	p -value
Age (years)	52.4 (9.2)	52.5 (10.1)	0.97
Height (cm)	156.6 (5.6)	156.0 (5.5)	0.70
Weight (kg)	54.0 (7.5)	52.7 (7.6)	0.60
Body mass index (kg/m^2)	22.1 (3.5)	21.7 (3.5)	0.77
Disease severity (no)			
Pre-osteoarthritis or initial stage	9	10	
Advanced or terminal stage	10	10	
Muscle strength (Nm/kg)			
Hip abductors	0.96 (0.33)	0.99 (0.29)	0.77
Hip extensors	1.15 (0.44)	1.21 (0.31)	0.60
Hip flexors	0.98 (0.35)	1.07 (0.32)	0.41
Knee extensors	1.64 (0.59)	1.65 (0.41)	0.98
Muscle power (W/kg)	4.11 (0.85)	4.24 (0.62)	0.56
Physical performance			
Maximum walking speed (m/s)	2.03 (0.41)	2.19 (0.37)	0.23
Timed Up and Go (s)	5.64 (1.18)	5.43 (1.04)	0.56
3-min walking test (m)	286.4 (60.0)	300.5 (40.4)	0.41
Clinical assessment			
Harris Hip Score (0–100)	75.4 (17.7)	77.2 (13.3)	0.72
Pain (0–100)	32.0 (24.9)	21.2 (18.0)	0.13
Muscle thickness (cm)			
Gluteus maximus	2.28 (0.49)	2.29 (0.37)	0.98
Gluteus medius	3.75 (0.74)	3.72 (0.52)	0.91
Quadriceps femoris	3.45 (0.65)	3.57 (0.61)	0.57
Muscle echo intensity (0–255)			
Gluteus maximus	91.4 (20.1)	89.1 (14.6)	0.70
Gluteus medius	85.9 (16.8)	82.7 (16.7)	0.55
Quadriceps femoris	94.9 (17.8)	91.6 (15.0)	0.53

Values are means (SD).

Discussion

Previous reports in older adults indicated a positive effect of high-velocity resistance training on muscle strength, power, and physical function.^{5–10} However, no studies have examined whether high-velocity training is more effective than low-velocity training for patients with hip osteoarthritis. The present study is the first to make this comparison. The primary findings of this study were a greater improvement in Timed Up and Go test in the high-velocity group than in the low-velocity group, while gains in hip and knee muscle strength in both groups. In

addition, the decrease in echo intensity of the gluteus maximus was greater in the high-velocity group than in the low-velocity group.

Muscle strengthening is recommended as a key component of most exercise regimens for hip and knee osteoarthritis. In this study, the extent of increases in isometric muscle strength were the same in both groups following the intervention. Contrary to our hypothesis, high-velocity and low-velocity training were equally effective in improving muscle power because no significant difference

Table 2. Effect of intervention on muscle function in the high-velocity and low-velocity groups.

Variables	High-velocity group		Low-velocity group		F-value	Difference between groups (95% confidence interval)
	Before	After	Changes	Before	After	Changes
Muscle strength (Nm/kg)						
Hip abductors	0.96 (0.33)	1.12 (0.40)	0.16 (0.14)	0.99 (0.29)	1.18 (0.24)	0.19 (0.18)
Hip extensors	1.15 (0.44)	1.34 (0.50)	0.19 (0.19)	1.21 (0.31)	1.44 (0.31)	0.22 (0.18)
Hip flexors	0.98 (0.35)	1.10 (0.37)	0.13 (0.16)	1.07 (0.32)	1.19 (0.33)	0.12 (0.15)
Knee extensors	1.64 (0.59)	1.79 (0.55)	0.15 (0.18)	1.65 (0.41)	1.93 (0.51)	0.29 (0.25)
Muscle power (W/kg)	4.11 (0.85)	4.36 (0.82)	0.25 (0.24)	4.24 (0.62)	4.36 (0.65)	0.12 (0.38)

Values are means (SD).

Table 3. Effect of intervention on physical performance and clinical assessment in the high-velocity and low-velocity groups.

Variables	High-velocity group		Low-velocity group		F-value	Difference between groups (95% confidence interval)
	Before	After	Changes	Before	After	Changes
Physical performance						
Maximum walking speed (m/s)	2.03 (0.41)	2.10 (0.43)	0.07 (0.16)	2.19 (0.37)	2.26 (0.36)	0.07 (0.16)
Timed Up and Go test (s)	5.64 (1.18)	5.18 (0.98)	-0.46 (0.27)	5.43 (1.04)	5.31 (0.79)	-0.23 (0.39)
3-min walking test (m)	286.4 (60.0)	302.7 (42.4)	16.4 (26.7)	300.5 (40.4)	313.3 (41.5)	11.6 (22.6)
Clinical assessment						
Harris Hip Score (0-100)	75.4 (17.7)	80.0 (15.5)	4.6 (10.5)	77.2 (13.3)	82.2 (13.4)	5.0 (11.2)
Pain (0-100)	32.0 (24.9)	24.7 (24.6)	-7.3 (21.7)	21.2 (1.80)	14.0 (20.6)	-7.2 (20.3)

Values are means (SD).

*p < 0.05: significant group difference adjusted for age, height, weight, and outcome measures before training.

Table 4. Effect of intervention on muscle property in the high-velocity and low-velocity groups.

Variables	High-velocity group			Low-velocity group			F-value	Difference between groups (95% confidence interval)
	Before		Changes	Before		Changes		
	After	After		After	After			
Muscle thickness (cm)								
Gluteus maximus	2.28 (0.49)	2.48 (0.53)	0.20 (0.37)	2.29 (0.37)	2.37 (0.44)	0.08 (0.23)	0.978	0.12 (−0.096, 0.276)
Gluteus medius	3.75 (0.74)	3.88 (0.64)	0.13 (0.30)	3.72 (0.52)	3.82 (0.58)	0.06 (0.48)	1.104	0.07 (−0.219, 0.687)
Quadriceps femoris	3.45 (0.65)	3.55 (0.63)	0.10 (0.18)	3.57 (0.61)	3.62 (0.61)	0.05 (0.28)	0.214	0.05 (−0.121, 0.192)
Muscle echo intensity (0–255)								
Gluteus maximus	91.4 (20.1)	84.4 (18.0)	−6.8 (9.0)	89.1 (14.6)	88.1 (13.3)	−1.0 (7.5)	5.264 *	−5.8 (−9.406, −0.547)
Gluteus medius	85.9 (16.8)	79.3 (15.4)	−6.7 (6.6)	82.7 (16.7)	81.3 (14.8)	−2.4 (9.2)	2.540	−4.3 (−7.847, 0.963)
Quadriceps femoris	94.9 (17.8)	91.4 (16.6)	−3.5 (8.9)	91.6 (15.0)	89.6 (15.0)	−2.0 (8.0)	0.108	−1.5 (−6.366, 4.595)

Values are means (SD).

* $p < 0.05$: significant group difference adjusted for age, height, weight, and outcome measures before training.

in change was found. Previous studies in healthy older adults reported that high-velocity training induces an increase in muscle power.^{5–7,9} However, other studies reported no difference in gain of muscle power between high-velocity and low-velocity training.^{8,28} Taken together, the results of this study may indicate that both types of training increase muscle power.

High-velocity training has been reported to improve physical performance beyond what can be achieved by traditional strength training.⁸ In this study, the decrease in the time for performing the Timed Up and Go test was greater in the high-velocity group (−0.46 s) than in the low-velocity group (−0.23 s), indicating that a more effective improvement in physical performance from high-velocity training for patients with hip osteoarthritis than from low-velocity training. Interestingly, the high-velocity group required less absolute total time than the low-velocity group during each exercise session, yet physical performance improved more in the high-velocity group than in the low-velocity group. However, no significant difference was found between the groups in levels of improvement in the other physical performance measures, maximum walking speed and 3-min walking test. The Timed Up and Go test includes multiple movements and high-velocity movement, such as standing up, walking, turning, walking back, and sitting down quickly. Therefore, the Timed Up and Go test may require more muscle power than walking ability, which is better measured by maximum walking speed and 3-min walking test. Thus, high-velocity training may not be more effective than low-velocity training for improvement of all physical performance measures, and our hypothesis was only partially supported.

The present study assessed physical disability using the Harris Hip Score scale, and found that the high- and low-velocity training similarly improved Harris Hip Score. The extent of decreases in pain measured on a visual analogue scale were the same following high- and low-velocity training. Previous studies in patients with hip and knee osteoarthritis reported improvements in physical disability and pain owing to

resistance training.^{11,29} Our results indicate no association of differences in movement velocity in resistance training with the extent of improvement in physical disability and pain of patients with hip osteoarthritis.

Unlike most previous studies, the present study assessed muscle properties, muscle thickness as an index of muscle mass, and echo intensity as an index of muscle composition. Hip osteoarthritis has been reported to cause changes in skeletal muscle properties, such as loss of muscle mass (muscle atrophy) and altered muscle composition including increased intramuscular adipose tissue.^{23,30} Previous studies demonstrated that resistance training led to an increase in muscle mass and a decrease in intramuscular fat tissue in the elderly.^{31,32} However, no study has investigated the effect of resistance training on these muscle properties in patients with hip osteoarthritis. Hence, the unique contribution of the present study is to compare the effects on muscle properties between high- and low-velocity training in patients with hip osteoarthritis. The results revealed that the extent of increase in the thickness of all muscles was the same following both types of training. However, a greater decrease in the echo intensity of the gluteus maximus was observed in the high-velocity group than in the low-velocity group. These results suggest that high-velocity training for patients with hip osteoarthritis effects positive changes in muscle composition, rather than changes in muscle mass. The results of this study showed that fat and fibrous tissues within the gluteus maximus may have decreased substantially following high-velocity training. One recent study demonstrated that ectopic fat cell formation in skeletal muscle was associated with action of the mesenchymal progenitors, which can be inhibited by satellite cells.³³ Mechanical stress owing to high-velocity contraction during high-velocity training may have caused activation of muscle satellite cells and subsequent inhibition of mesenchymal progenitors. However, quantitative studies in support of this hypothesis are scarce; thus, future studies are needed. Physiological evidence suggests that

muscle composition as well as muscle mass are important elements in generating muscle contraction and physical function.^{16,34} The improvement in physical performance observed in the high-velocity group may therefore have been caused by changes in muscle composition.

The results of this study revealed that the effect of high-velocity training for patients with hip osteoarthritis was limited to the Timed Up and Go test and muscle composition of the gluteus maximus, among a range of outcome measurements, when compared with the effect of low-velocity training. Nevertheless, these results have important implications for the future design of training programmes for patients with hip osteoarthritis, especially as changes resulting from high-velocity training occurred in less total time per exercise session. However, two participants in the high-velocity group discontinued the exercise programme because of increased hip pain. Thus, high-velocity training for patients with hip osteoarthritis may not always be safer and may increase the risk of hip pain, even at low resistance.

This study has several limitations. First, the number of study participants was small, and the drop-out rate was relatively high. The small sample size may have resulted in type 2 errors. Second, the outcome assessors and the physical therapist who prescribed the intervention programme were not blinded to group allocation, which may have led to bias in the results. Third, the compliance rate relied on participants' completion of the training diary, and the exercise load was determined subjectively. However, the study examiners made every effort to confirm the compliance and exercise load during the intervention by telephone confirmation every two weeks.

In conclusion, eight weeks of high-velocity training at low resistance for patients with hip osteoarthritis had partially a greater effect on muscle composition and physical performance compared with low-velocity training. Although further study is needed, our results have important implications for the future design of training programmes for safe and effective enhancement of physical function in patients with hip osteoarthritis.

Clinical messages

- Home-based resistance training at high velocity for patients with hip osteoarthritis has a slightly greater effect on physical performance and muscle composition than low-velocity training.
- High-velocity training is an efficient intervention programme for patients with hip osteoarthritis because it required lesser absolute total time per exercise session than low-velocity training.

Acknowledgements

The authors express their thanks to K. Goto (Department of Orthopaedic Surgery, Nagahama City Hospital, Shiga, Japan) and Y. Yamada (Graduate School of Nursing, Kyoto Prefectural University of Medicine, Kyoto, Japan) for their helpful suggestions as well as K. Uemura (Graduate School of Medicine, Nagoya University, Aichi, Japan) and A. Matsumura (Graduate School of Medicine, Kyoto University, Kyoto, Japan) for their practical technical assistance.

Conflict of interest

The authors declare that there is no conflict of interest.

Funding

This study was supported by a research fellowship award to FY from the Japan Society for the Promotion of Science for Young Scientists (22-7453).

References

1. Zhang W, Nuki G, Moskowitz RW, et al. OARSI recommendations for the management of hip and knee osteoarthritis: part III: Changes in evidence following systematic cumulative update of research published through January 2009. *Osteoarthritis Cartilage* 2010; 18: 476–499.
2. Zhang W, Doherty M, Arden N, et al. EULAR evidence based recommendations for the management of hip osteoarthritis: report of a task force of the EULAR Standing Committee for International Clinical Studies Including Therapeutics (ESCISIT). *Ann Rheum Dis* 2005; 64: 669–681.
3. Suzuki T, Bean JF and Fielding RA. Muscle power of the ankle flexors predicts functional performance in community-dwelling older women. *J Am Geriatr Soc* 2001; 49: 1161–1167.
4. Cuoco A, Callahan DM, Sayers S, Frontera WR, Bean J and Fielding RA. Impact of muscle power and force on gait speed in disabled older men and women. *J Gerontol A Biol Sci Med Sci* 2004; 59: 1200–1206.
5. Hakkinen K, Alen M, Kallinen M, Newton RU and Kraemer WJ. Neuromuscular adaptation during prolonged strength training, detraining and re-strength-training in middle-aged and elderly people. *Eur J Appl Physiol* 2000; 83: 51–62.
6. Hakkinen K, Kallinen M, Izquierdo M, et al. Changes in agonist-antagonist EMG, muscle CSA, and force during strength training in middle-aged and older people. *J Appl Physiol* 1998; 84: 1341–9.
7. Fielding RA, LeBrasseur NK, Cuoco A, Bean J, Mizer K and Fiatarone Singh MA. High-velocity resistance training increases skeletal muscle peak power in older women. *J Am Geriatr Soc* 2002; 50: 655–662.
8. Miszko TA, Cress ME, Slade JM, Covey CJ, Agrawal SK and Doerr CE. Effect of strength and power training on physical function in community-dwelling older adults. *J Gerontol A Biol Sci Med Sci* 2003; 58: 171–175.
9. Earles DR, Judge JO and Gunnarsson OT. Velocity training induces power-specific adaptations in highly functioning older adults. *Arch Phys Med Rehabil* 2001; 82: 872–878.
10. Orr R, de Vos NJ, Singh NA, Ross DA, Stavrinou TM and Fiatarone-Singh MA. Power training improves balance in healthy older adults. *J Gerontol A Biol Sci Med Sci* 2006; 61: 78–85.
11. van Baar ME, Dekker J, Oostendorp RA, et al. The effectiveness of exercise therapy in patients with osteoarthritis of the hip or knee: a randomized clinical trial. *J Rheumatol* 1998; 25: 2432–2439.
12. Kettunen JA and Kujala UM. Exercise therapy for people with rheumatoid arthritis and osteoarthritis. *Scand J Med Sci Sports* 2004; 14: 138–142.
13. Rydevik K, Fernandes L, Nordsletten L and Risberg MA. Functioning and disability in patients with hip osteoarthritis with mild to moderate pain. *J Orthop Sports Phys Ther* 2010; 40: 616–624.
14. Takatori Y, Ito K, Sofue M, et al. Analysis of interobserver reliability for radiographic staging of coxarthrosis and indexes of acetabular dysplasia: a preliminary study. *J Orthop Sci* 2010; 15: 14–19.
15. Tsukagoshi R, Tateuchi H, Fukumoto Y, Okumura H and Ichihashi N. Stepping exercises improve muscle strength in the early postoperative phase after total hip arthroplasty: a retrospective study. *Am J Phys Med Rehabil* 2012; 91: 43–52.
16. Fukumoto Y, Ikezoe T, Yamada Y, et al. Skeletal muscle quality assessed from echo intensity is associated with muscle strength of middle-aged and elderly persons. *Eur J Appl Physiol* 2012; 112: 1519–1525.
17. Suetta C, Andersen JL, Dalgaard U, et al. Resistance training induces qualitative changes in muscle morphology, muscle architecture, and muscle function in elderly postoperative patients. *J Appl Physiol* 2008; 105: 180–186.

18. Bohannon RW. Comfortable and maximum walking speed of adults aged 20–79 years: Reference values and determinants. *Age Ageing* 1997; 26: 15–19.
19. Podsiadlo D and Richardson S. The Timed up and Go - a Test of Basic Functional Mobility for Frail Elderly Persons. *J Am Geriatrics Society* 1991; 39: 142–148.
20. Iriberrri M, Galdiz JB, Gorostiza A, Ansola P and Jaca C. Comparison of the distances covered during 3 and 6 min walking test. *Respir Med* 2002; 96: 812–816.
21. Soderman P and Malchau H. Is the Harris hip score system useful to study the outcome of total hip replacement? *Clin Orthop Relat R* 2001; 384: 189–197.
22. Ikezoe T, Mori N, Nakamura M and Ichihashi N. Atrophy of the lower limbs in elderly women: is it related to walking ability? *Eur J Appl Physiol* 2011; 111: 989–995.
23. Fukumoto Y, Ikezoe T, Tateuchi H, et al. Muscle mass and composition of the hip, thigh and abdominal muscles in women with and without hip osteoarthritis. *Ultrasound Med Biol* 2012; 38: 1540–1545.
24. Pillen S, Tak RO, Zwarts MJ, et al. Skeletal Muscle Ultrasound: Correlation between Fibrous Tissue and Echo Intensity. *Ultrasound Med Biol* 2009; 35: 443–446.
25. Sipila S and Suominen H. Muscle ultrasonography and computed tomography in elderly trained and untrained women. *Muscle Nerve* 1993; 16: 294–300.
26. Dupont AC, Sauerbrei EE, Fenton PV, Shragge PC, Loeb GE and Richmond FJ. Real-time sonography to estimate muscle thickness: comparison with MRI and CT. *J Clin Ultrasound* 2001; 29: 230–236.
27. Miyatani M, Kanehisa H, Ito M, Kawakami Y and Fukunaga T. The accuracy of volume estimates using ultrasound muscle thickness measurements in different muscle groups. *Eur J Appl Physiol* 2004; 91: 264–272.
28. Henwood TR, Riek S and Taaffe DR. Strength versus muscle power-specific resistance training in community-dwelling older adults. *J Gerontol A Biol Sci Med Sci* 2008; 63: 83–91.
29. Jan MH, Lin JJ, Liao JJ, Lin YF and Lin DH. Investigation of clinical effects of high- and low-resistance training for patients with knee osteoarthritis: a randomized controlled trial. *Phys Ther* 2008; 88: 427–436.
30. Rasch A, Bystrom AH, Dalen N and Berg HE. Reduced muscle radiological density, cross-sectional area, and strength of major hip and knee muscles in 22 patients with hip osteoarthritis. *Acta Orthop* 2007; 78: 505–510.
31. Taaffe DR, Henwood TR, Nalls MA, Walker DG, Lang TF and Harris TB. Alterations in muscle attenuation following detraining and retraining in resistance-trained older adults. *Gerontology* 2009; 55: 217–223.
32. Sipila S and Suominen H. Quantitative ultrasonography of muscle: detection of adaptations to training in elderly women. *Arch Phys Med Rehabil* 1996; 77: 1173–1178.
33. Uezumi A, Fukada S, Yamamoto N, Takeda S and Tsuchida K. Mesenchymal progenitors distinct from satellite cells contribute to ectopic fat cell formation in skeletal muscle. *Nat Cell Biol* 2010; 12: 143–152.
34. Visser M, Kritchevsky SB, Goodpaster BH, et al. Leg muscle mass and composition in relation to lower extremity performance in men and women aged 70 to 79: the health, aging and body composition study. *J Am Geriatr Soc* 2002; 50: 897–904.